Light's Criteria — Pleural Effusion (Exudate vs Transudate)

diagnostic pleural tap with paired serum · ward

Light's

EXUDATE IF ANY ONE IS TRUE

Protein ratio: `pleural protein ÷ serum protein > 0.5`
LDH ratio: `pleural LDH ÷ serum LDH > 0.6`
Pleural LDH: `> 2/3 × upper limit of normal serum LDH`

RESULT

Any criterion met Exudate

None met Transudate

COMMON CAUSES

Transudate	Exudate
Heart failure	TB (lymphocytic) — most common in SA
Cirrhosis (hepatic hydrothorax)	Parapneumonic effusion / empyema
Nephrotic syndrome, hypoalbuminaemia	Malignancy
Peritoneal dialysis	PE, pancreatitis, connective tissue disease

SA STG

- Tube drainage if pleural pH <7.2, bacteria on Gram stain/culture, or frank pus ✚
- Lymphocytic exudate in a high-TB-burden area → empiric TB treatment ✚
- Drain max 1 L at a time; TB empyema needs an intercostal drain ✚
- Children: Light's criteria (calculator) for pleural fluid ✚

PEARLS

- Send pleural protein, LDH, glucose, pH, cell count & differential, Gram/AFB/GeneXpert, ADA; serum protein + LDH same time
- Light's misclassifies ~25% of diuretic-treated heart failure as exudate — serum–pleural albumin gradient >12 g/L favours transudate
- Sensitivity ≈98% for exudate

HOSP STG & EML Hospital Level Adults, 6th ed. 2024 (updated Mar 2026) — pp 16.18, 16.21 · **PAED** STG & EML Paediatric Hospital Level, 2025-2029 cycle (updated Mar 2026) — pp 15.13 · **LIGHT** Light RW et al. Ann Intern Med 1972;77:507-13